

A multi-level model for preferences

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Received 15 September 2000; received in revised form 4 December 2000; accepted 22 December 2000

Abstract

The study of preferences on different levels of aggregation is very important. In this paper measurements of preferences are performed as paired comparisons and a multi-level model of preferences is presented. Within a multilevel model it is possible to give an exact interpretation of aggregated preferences as conditional expected preferences. Furthermore, hypotheses can be stated in an exact way and the statistical analysis can be carried out as an ANOVA. © 2001 Elsevier Science Ltd. All rights reserved.

Keywords: Aggregated preferences; Analysis of variance; Individual preferences; Paired comparisons

1. Introduction

Within the social sciences and especially within the theory of consumer behaviour the concepts of preferences and choice behaviour are common. Preferences may be assigned to individuals, different subgroups or to a population.

Furthermore, for a set of objects it might be assumed that the preferences for any pair of objects can be related to the difference between the two objects on a continuous scale i.e. the objects can be arranged on a rating scale. In this case we say that there exist a preference function.

In practical applications it is often desirable to consider preferences and preference functions on different levels of aggregation. The lowest level may be the level of subjects, i.e. individual preference functions. An intermediate level may be individual preference functions aggregated to the level of e.g. gender, in which case you talk about a preference function for men and a preference function for women, these preference functions possibly being different. The preference functions may be further aggregated to the highest level, the population, in which case one would talk about a preference function for the whole population.

Also, concerning formulation and testing of hypotheses more than one level may be of interest. On the level

of individuals one may want to test the hypothesis that all men have identical preference functions and similarly that women have identical preference functions. However, on the level of gender one may want to test the hypothesis that men and women have identical preference functions.

The method of paired comparisons is a basic tool for measuring preferences for stimuli, as well as for scaling responses to objects or perceptual attributes (David, 1988). The power of this method lies in its ability to make sharp discrimination of scaled objects. The major parts of scaling models for this kind of data are probabilistic, dating back to Thurstone (1927). For graded paired comparisons an analysis-of-variance model at the aggregated level was first developed by Scheffé (1952). This approach was extended to the individual level by Bechtel (1967, 1976) and further extended to permit any (unknown) pattern of correlations among the paired comparisons. Also, explanatory categorical variables of subjects as well as of objects can be included in the analysis-of-variance (Gabrielsen, 2000).

In the present paper a model for graded paired comparisons is developed which simultaneously comprises several levels. This is achieved by construction of a so-called multi-level model or a random effect model. The idea is to consider some parameters as random variables such that preference functions at different levels of the model appear as conditional expected preference functions after conditioning on parameters related to different levels.

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From a mathematical point of view, the construction of a random effect model of preference functions is rather complicated. If, however, the experiment is carefully designed the practical application of the model is straightforward. This holds for estimation and testing as well as for interpretation. As with testing of hypotheses the usual Analysis of Variance Table (ANOVA) can be applied.

The aim of the paper is to demonstrate the capabilities of a multilevel model. The approach is illustrated by an example in which hypotheses and analyses are discussed. The experiment discussed was performed as a pilot study to evaluate the feasibility of the experiment, the reasonableness of the statistical analyses and the interpretation of consequent results.

2. Aim of the experiment

Prior to the elaboration of an advertising campaign for a specific Danish beer it was decided to examine differences in preferences to beer in Denmark. Initially, it was assumed that men had higher preference for the specific beer than women. If this was the case the campaign should be designed to take this difference into account. Furthermore, it was of importance to know how homogeneous men (and women) were in preferences. If there was a high agreement among men (women) the presentation could be more specific, if the agreement was low the presentation should be more general.

Therefore, the experiment was performed to test a hypothesis of equal preference functions for men and women and, furthermore, to obtain a measure of the agreement of preferences among men and similarly for women.

3. The experiment

To estimate the preferences five other common Danish beers were chosen such that the total of the six beers covered the majority of the Danish market of standard beers. In this way the preference for the specific beer could be determined relatively to the six beers comprising the Danish standard beer market. Fifty students in marketing classes at the Copenhagen Business School were randomly chosen to take part in the experiment, 25 men and 25 women.

As one of the objects of the experiment was to obtain measures of homogeneity of preferences within and between genders it was decided to perform the experiment as paired comparisons. Each subject was placed at a computer screen showing two pictures of beers, one beer to the left and one to the right. Subjects were instructed to rate their preference (electronically) on a

30 cm linear scale running below the two pictures. The scale was labelled at left anchor “I strongly prefer left beer” and at the right anchor “I strongly prefer right beer”. In the middle the scale was labelled “I have no preference”. After the preference was rated the subject pressed a bottom marked “continue” and a new screen with two new pictures of beer appeared. This was repeated until 12 comparisons were performed. To ensure the proper understanding of the comparisons three additional warm-up comparisons were performed as an opening. For each subject the whole experiment lasted 4–5 min.

At the end of the experiment ratings were converted into numbers. The distance from the middle of the scale to the noted mark was measured (electronically), positively to the right and negatively to the left. The corresponding value of the score of subject p with beer i to the left and beer j to the right is denoted y_{ijp} . In any event it was assumed that the numerical score increases with the strength of the preference for j over i , and that equal but opposite preferences (j over i , and i over j) corresponds to equal but opposite scores. Thus, for subject p

y_{ijp} is the assessed preference of beer j over beer i

and

y_{jip} is the assessed preference of beer i over beer j .

If, e.g. y_{ijp} , is positive j is preferred over i , if y_{ijp} is negative i is preferred over j .

The number of comparisons each subject can be exposed to is limited if the reliability of the comparisons is to be sustained. Denoting the six beers as 1, ..., 6 a comparison of beer i and j , with beer i as the left picture and beer j as the right picture ($i \neq j$) can be denoted (i, j) . For the present experiment we chose among the 30 possible comparisons the following twelve: (1,2)(2,1), (2,3)(3,2), (3,4)(4,3), (4,5)(5,4), (5,6)(6,5), (6,1) and (1,6). The inclusion of six “replications”, e.g. (1,2) and (2,1), permits the estimation of the measurement error, thereby making it possible to distinguish measurement error from variation due to deviation of preferences from preference functions. Furthermore, it is possible to test the hypothesis of preference functions, Section 5. The applied “design” is not balanced, however, this just implies that all preferences between two beers are not estimated with the same precision or variance, which is of less importance in the present case.

4. The model

The underlying assumptions of the mathematical model are that all the y_{ijp} are independent random variables and normally distributed with mean value $E(y_{ijp}) = \xi_{ijp}$ and the same variance, that is

$$y_{ijp} \sim N(\xi_{ijp}, \sigma^2). \quad (1)$$

The mean-values may be re-written $\xi_{ijp} = \theta_{ijp} + \tau_{ijp}$, where for all i, j and p $\theta_{ijp} = -\theta_{jip}$ and $\tau_{ijp} = \tau_{jip}$. θ_{ijp} is the ‘true’ preference of beer j over beer i and it is assumed that the preferences are anti-symmetric ($\theta_{ijp} = -\theta_{jip}$). τ_{ijp} is the order effect of subject p comparing beer i and beer j . To obtain degrees of freedom to estimate the residual variance a reasonable assumption is that τ_{ijp} depend on the subjects but not on the pair of beers to be compared. Thus, the mean value is specified as

$$\xi_{ijp} = \theta_{ijp} + \tau_p. \quad (2)$$

If τ_p is positive, subject p has a bias towards a preference of the beer to the right, and conversely if τ_p is negative. From the point of view of ANOVA the order effects, τ_p , are subject-main-effects, however, in the present context the interpretation of τ_p is a systematic bias of subject p measured as deviation from zero.

Model (1) — and all subsequent considered models — falls under the general theory of least squares and linear hypotheses and may be written in the usual way as

$$y = X\beta + \varepsilon \quad (3)$$

for an appropriate design matrix, X . A detailed discussion of this model is found in Gabrielsen (2000), which also gives a detailed description of how to construct the necessary design matrices and to perform the calculations.

4.1. Checking model assumptions

The normality assumption is not likely to be a problem, do to the robustness of ANOVA. However, high preponderance of data at the end points of the scale may cause problems. This problem was anticipated by using the labels “...strongly prefer...” at the anchors to avoid subjects approaching the end points of the scale. Subsequent analysis of the raw data showed no “jamming” at the end point. Also, probit-diagrams of standardised residuals showed no irregularities.

The analysis-of-variance is sensitive to violation of the homogeneity of variance assumption, however, this assumption can be statistically tested. Model (1) is easily extended to allow for subject dependent variances

$$y_{ijp} \sim N(\theta_{ijp} + \tau_p, \sigma_p^2) \quad (4)$$

For each subject seven mean value parameters are estimated leaving 5 degrees of freedom to estimate a residual variance. The hypothesis of homogeneity of variances, thus, can be tested using Bartlett’s test which gives $\chi^2 = 38.96$, D.F. = 49, $P = 0.8473$. Therefore, the hypothesis of homogeneity of variances can be

maintained. In the majority of practical applications the hypothesis is rejected. However, the analysis-of-variance can still be performed, see Bechtel, Ofir, and Khuri (1995).

For individual paired comparisons the assumption of independence of observation may be violated. For a further discussion of this case see Bechtel et al. (1995).

5. Statistical hypotheses and analysis of variance

Each subject performed 12 comparisons such that the total number of observations is 600. For each subject Model (2) includes six preferences, each measured twice but in opposite directions, plus one order effect. Having 50 subjects a total of 350 parameters are estimated in Model (2).

Since the measurements are preferences or distances the hypothesis that all mean-values are zero has the specific interpretation that no preferences exist. Therefore, the constant is included in the total variation, which is calculated as $\sum y_{ijp}^2$ having 600 degrees of freedom. To identify significant effects the mean-value specification is de-composed by specifying series of mean-value hypotheses.

Initially, the effect of the total model can be decomposed into an effect of true preferences (θ_{ijp}) and an effect of order effects (τ_p), Table 1. These two effects can again be decomposed.

5.1. Hypotheses of order effects

Specifying a hypothesis of common order effect for all subjects

$$H : \tau_p = \tau$$

and subsequently a hypothesis of vanishing common order effect

$$H : \tau_p = \tau = 0$$

decomposes the order effects into a between subjects order effect and a common order effect (Table 2).

Table 1
Analysis of variance. Model (2)

Effect	d.f.	s.s.	m.s.	F	P
Order effects	50	93.6	1.872	2.037	<0.001
Preferences	300	15 432.2	51.441	55.975	<0.001
Model (2)	350	15 525.8	44.359	48.269	<0.001
Residual	250	229.8	0.919		
Total	600	15 755.6	26.259		

5.2. Hypotheses of preferences

An important hypothesis states that for each subject the assessed preference for any pair of beers can be attributed to the difference of the two beers on a continuous scale, i.e. the beers can be arranged on a rating scale. One way of expressing this hypothesis is that for each subject there exists α 's corresponding to the objects so that

$$H_1 : \theta_{ijp} = \alpha_{jp} - \alpha_{ip}, \quad p = 1, \dots, 50, \quad i, j = 1, \dots, 6 \quad (i \neq j). \quad (5)$$

As only the differences of α 's can be estimated one can add the restriction that for each subject the α 's should sum to zero. If such a rating scale exists it is called a preference function. Thus, H_1 states the existence of individual preference functions.

If H_1 can be supported a subsequent hypothesis is whether the preference functions are identical for all men and, similarly, identical for all women, these preference functions possibly being different. This hypothesis states

$$H_2 : \alpha_{ip} = \alpha_{i, \text{gender}(p)}, \quad (6)$$

where $\text{gender}(p)$ is the gender of subject p .

If H_2 is supported an important subsequent hypothesis is whether the common preference function for men and the common preference function for women are equal, that is that all subjects have the same preference function

$$H_3 : \alpha_{ip} = \alpha_i$$

The hypothesis H_3 states that there is a common preference function for the whole population.

Finally, a further subsequent hypothesis is that there are no preferences, the objects are equally well preferred. This hypothesis becomes

$$H_4 : \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = \alpha_6.$$

The subsequent hypotheses concerning preferences correspond to an orthogonal decomposition such that the analysis of variance table for preferences becomes Table 3.

The series of subsequent hypothesis concerning preferences (θ 's) are collected in Table 4, giving also the number of parameters to be estimated corresponding to the dimension of the mean-value-space.

The three tables Tables 1, 2 and 3 are all based on orthogonal decompositions and can of course be combined into a single analysis of variance table taking in all of the effects, the decomposed order effects, the decomposed preference effects and the residual. The advantage of using orthogonal decompositions is that the effects can be estimated independently of each other. This means that removing some of the effects does not affect the estimates of the remaining effects.

5.3. Analysis of variance

The residual sum of square is estimated to 229.8 with 250 degrees of freedom giving an estimated residual variance 0.919 (Table 1). For the sake of simplicity this residual variance is applied as denominator in all the following F -tests, as long as fixed-effect models are considered.

Table 2
Analysis of variance. Order effects

Effect	d.f.	s.s.	m.s.	F	P
Different order effects	49	45.5	0.929	1.011	0.461
Common order effect	1	48.1	48.100	52.339	< 0.001
Total	50	93.6	1.872	2.037	< 0.001

Table 3
Analysis of variance. Preferences

Effect	d.f.	s.s.	m.s.	F	P
Deviation from preference functions	50	61.8	1.236	1.345	0.075
Within gender variation of preference functions	240	4805.4	20.023	21.788	< 0.001
Between gender variation of preference functions	5	137.0	27.400	29.815	< 0.001
Common preference functions	5	10 428.0	2085.600	2269.423	< 0.001
Total	300	15 432.2	51.441	55.975	< 0.001

Table 4
Subsequent hypotheses concerning preferences

Hypothesis	No. of parameters
Model (2) ($\theta_{ijp} = -\theta_{jip}$)	300
$H_1 : \theta_{ijp} = \alpha_{jp} - \alpha_{ip}$	250
$H_2 : \alpha_{ip} = \alpha_{i, \text{gender}(p)}$	10
$H_3 : \alpha_{ip} = \alpha_i$	5
$H_4 : \alpha_i = 0$	0

Concerning the order effects (Table 2), these are the same for all subjects ($F=1.011$, $P=0.461$) but the common order effect is highly significant, ($F=52.339$, $P\leq 0.001$). The common order effect is estimated to $\hat{\tau}=0.28$ cm. The interpretation of this common order effect is that the labelling of the middle of the scale “I have no preference” by a technical error was displaced slightly to the right and thereby introduced a bias in subjects visual perception of the middle of the scale. This also explains that the order effects were identical for all subjects since the bias was due to the set up of the experiment and thereby influenced all subjects in the same way. To some extent this finding validates the experiment.

However, the presence of this common order effect does not give rise to any bias in the results of the experiment concerning preferences. This is due to the fact that the experiment is designed in such a way that possible order effects are orthogonal to the effect of preferences and therefore do not influence the effects and estimates of preferences.

Concerning the preferences it is found that the model has explanatory power ($F=55.975$, $P<0.001$), Table 1, and the total effect of preferences is, therefore, further decomposed.

From Table 3 it is seen that subjects are reasonable consistent in their preferences so that H_1 is supported ($F=1.345$, $P=0.075$), and it can be assumed that there exist individual preference functions.

The hypothesis of within gender homogeneity of preference function is rejected ($F=21.788$, $P<0.001$).

The interpretation of this is that the variation of individual preference functions is so high that there is no indication of a common preference function for men, and a common preference function for women. Neither is there any indication of a common preference function for the whole population.

One of the original aims of the experiment was to examine whether the preference functions for men and women are equal. However, this hypothesis cannot be tested, and in the fixed parameter model the hypothesis cannot even be formulated due to the large differences in preference functions for men and similarly for women. This is obviously an unsatisfactory result and an alternative model is therefore constructed to overcome this problem.

6. A multilevel model

The basic idea of a multilevel model is to consider certain parameters as stochastic variables. In the present case we are only interested in hypotheses concerning preferences. The order effects can be considered as nuisance parameters and since the decomposition is ortho-

gonal inclusion of the order effects will have no influence on estimates of preferences. In addition to the residual variance σ^2 we introduce a component of variance corresponding to each of the hypotheses concerning preferences, i.e. H_1 , H_2 , H_3 and H_4 .

As a first step we consider θ_{ijp} as normally distributed random variables with expected value $E(\theta_{ijp}) = \alpha_{jp} - \alpha_{ip}$ and variance σ_1^2 . In this way the deviation of preferences from preference functions is considered as random and the variance, σ_1^2 , is a measure of this deviation. For an exact definition of variance component models see Appendix.

To be able to talk about preference functions for each gender we consider the α_{ip} 's as random variables with expected value $E(\alpha_{ip}) = \alpha_{i,\text{gender}(p)}$ and variance σ_2^2 so that σ_2^2 is a measure of the within gender variation of preference functions.

To be able to talk about a common preference function for the whole population we consider $\alpha_{i,\text{gender}(p)}$ as random variables with expected value $E(\alpha_{i,\text{gender}(p)}) = \alpha_i$ and variance σ_3^2 so that σ_3^2 is a measure of the between gender variation of preference functions.

Finally we can consider the α_i 's as random variables with expected value zero and variance σ_4^2 . This variance, σ_4^2 , becomes a measure of the differences in preferences expressed by the common preference function.

6.1. Interpretation

The interpretation of the multilevel model can be illustrated by comparing the multi-level model to the fixed-effect model.

In the fixed-effect model $H_1 : \theta_{ijp} = \alpha_{jp} - \alpha_{ip}$ states the existence of individual preference functions and if H_1 is rejected individual preference functions do not exist. In the multi-level model $\theta_{ijp} = \alpha_{jp} - \alpha_{ip}$ can be interpreted as the expected preference functions conditioning on the subjects and σ_1^2 is the variance in the distribution describing the deviance of preferences from preference functions.

In the fixed-effect model $H_2 : \alpha_{ip} = \alpha_{i,\text{gender}(p)}$ states the existence of preference functions counting for each gender. If H_2 is rejected no such preference functions exist. In the multi-level model $\alpha_{ip} = \alpha_{i,\text{gender}(p)}$ can be interpreted as the expected preference function conditioning on the gender of subjects and σ_2^2 is the variance in the distribution describing the within gender variation of preference functions.

In the fixed-effect model $H_3 : \alpha_{ip} = \alpha_i$ states the existence of a common preference function for the whole population. If H_3 is rejected no such preference function exists. In the multi-level model $\alpha_{ip} = \alpha_i$ can be interpreted as the expected preference function conditioning on $\alpha_1, \dots, \alpha_6$ and σ_3^2 is the variance in the distribution describing the between gender variation of preference functions.

Finally, and maybe of less interest, is $H_4 : \alpha_1 = \dots = \alpha_6$ the hypothesis of no preferences, i.e. the objects are alike with respect to preferences. In the multilevel model $\alpha_1 = \dots = \alpha_6$ is the expected (degenerated) preference function and σ_4^2 is the variance in the distribution describing the variation of preferences. A small variance indicates that the beers are of almost the same preference, a high variance indicates that preferences to beers are very different.

In a multi-level model, therefore, preference functions on different levels can be interpreted as expected preference functions conditioning on relevant variables related to different levels. In the present case three levels are relevant, namely the level of individuals, the level of gender and the level of the population. The expected preference functions related to the three levels are obtained by conditioning on subjects, gender and no conditioning, respectively.

6.2. Hypotheses

The hypotheses considered in the fixed-effect model can also be considered in the multi-level model.

The hypothesis $H_1 : \theta_{ijp} = \alpha_{jp} - \alpha_{ip}$ corresponds in the multi-level model to the hypothesis $H_1 : \sigma_1^2 = 0$. Note, that if $\sigma_1^2 = 0$, the expected value of the m.s. corresponding to deviation from individual preference

functions is σ^2 , which is equal to the expected value of the m.s. corresponding to the residual. Therefore, the two s.s. may be summed to obtain a pooled estimate of σ^2 .

Similarly, the hypothesis $H_2 : \alpha_{ip} = \alpha_{i,\text{gender}(p)}$ corresponds to $H_2 : \sigma_2^2 = 0$. The hypothesis $H_3 : \alpha_{ip} = \alpha_i$ corresponds to $H_3 : \sigma_3^2 = 0$ and finally $H_4 : \alpha_1 = \dots = \alpha_6$ corresponds to $H_4 : \sigma_4^2 = 0$.

6.3. Estimation

The estimates of conditional preference functions in the multi-level model are exactly the same as the corresponding mean-value estimates in the fixed-effect model.

The estimated preference functions at the level of gender, i.e. separate preference functions for men and women are shown in Table 5, which also shows the estimated common preference function at the level of the population. It is seen that men to a little degree prefer the specific beer, which is number 1, higher than women do.

If all levels of preferences are considered as random the variance components becomes $\sigma^2, \sigma_1^2, \dots, \sigma_4^2$. The estimation is performed by successively solving the equations given in Table 6. For the present data the solution is shown in Table 7. Note, that by having 'replications' included in the paired comparisons it is possible to estimate both σ^2 and σ_1^2 . Otherwise, these variance components would have been confounded.

The residual variance is estimated to $\hat{\sigma}^2 = 0.919$ as in the fixed-effect model, Table 1. The variance component corresponding to deviation from individual preference functions is small, $\hat{\sigma}_1^2 = 0.317$, compared to the residual variance. The within gender variation of preference functions is high, $\hat{\sigma}_2^2 = 18.787$. Furthermore, the between gender variation of preference functions is high

Table 5
Estimated preference functions

	α_5	α_4	α_6	α_1	α_3	α_2
Women	-3.70	-3.03	-1.90	-1.59	3.87	6.35
Men	-4.06	-2.58	-1.90	-0.77	3.86	5.44
Common	-3.88	-2.80	-1.90	-1.18	3.87	5.90

Table 6
Expected and estimated effects

Effect	Expected m.s.	Estimated m.s.
Residual	σ^2	0.919
Deviation from preference functions	$\sigma^2 + \sigma_1^2$	1.236
Within gender variation of preference functions	$\sigma^2 + \sigma_1^2 + \sigma_2^2$	20.023
Between gender variation of preference functions	$\sigma^2 + \sigma_1^2 + \sigma_2^2 + \sigma_3^2$	27.400
Common preference function	$\sigma^2 + \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2$	2085.600

Table 7
Estimated variance components

Effect	Variance component	Estimate
Residual	σ^2	0.919
Deviation from individual preference function	σ_1^2	0.317
Within gender variation of preference function	σ_2^2	18.787
Between gender variation of preference functions	σ_3^2	7.377
Variation of common order effect	σ_4^2	2058.200

$\hat{\sigma}_3^2 = 7.377$, however, smaller than the within gender variation. This indicates that the variation of preference functions among men (and similarly among women) is so high that it has little meaning to talk about differences in preference functions for men and women, respectively.

6.4. Test of hypotheses

The hypotheses in the fixed-effect model are tested by usual F -tests. The tests in the multi-level model are identical and the approach therefore can be taken in the analysis of variance table, which is identical for the two models. It is sometimes argued that the F -test in multi-level models should be two-tailed. For a discussion of this matter see Tjur (1984).

An F -test shows that the hypothesis $\sigma_1^2 = 0$ can be supported, $P = 0.075$, Table 3.

An important difference in the two models is that if the hypothesis of within gender homogeneity of preference functions is rejected the analysis comes to a stop in the fixed effect model. In the multilevel model, on the contrary, the hypothesis of identical preference functions for men and women can still be formulated as expected preference function. However, as H_2 is rejected the variances cannot be pooled and the hypothesis H_3 must be tested against the s.s. corresponding to within gender variation. In the present case we find $F(5,240) = 27.400/20.023 = 1.368$, $P = 0.237$, such that the hypothesis of identical preference functions for men and women cannot be rejected. This reflects the fact that the variance component corresponding to between gender variation, σ_3^2 , is small compared to the within gender variation, σ_2^2 , Table 7. Expressed in another way — the difference between the estimated preference functions for men and women (Table 5), is small compared with the within gender variation of preference functions.

Note that if the pooled variance has been used, H_3 had been rejected but this test is in some sense illegal since H_2 is rejected.

7. Result of experiment

The between gender variation of preference functions was found to be high, $\hat{\sigma}_3^2 = 7.377$, compared with the residual variation, which was estimated to $\hat{\sigma}^2 = 0.919$. This indicates that there might be a difference in preferences according to gender. An examination of the estimated preference functions, Table 5, supports that an average man compared with an average woman prefers the specific beer to some degree.

However, the within gender variation of preference functions was estimated to $\hat{\sigma}_2^2 = 18.787$, which indicates a very high degree of within gender variation compared with the between gender variation. The within gender

preferences, therefore, are very inhomogeneous. Compared with this inhomogeneity the between gender differences in preference functions can be neglected.

The conclusion of the experiment, therefore, is that the advertising campaign should be presented in a very general way without any regard to gender.

8. Conclusion

Multi-level models can in many situations be an attractive alternative to fixed-effect models. A careful design of the experiment implies that the analysis of variance table can be used to test hypotheses.

A multi-level model has several advantages compared to a usual fixed effect model.

1. Hypotheses specified in the fixed effect model can also be formulated in the random effect model, and tested in a similar way. The random effect model, therefore, can be considered as an extension of the fixed effect model.
2. Aggregation and disaggregation is performed within the model such that it is possible to study how effects appearing at lower levels by aggregation is transformed to higher levels.
3. If, e.g. the hypothesis of homogeneity of preference functions within groups (e.g. defined by gender), is rejected there is no interpretation of group specific preference functions in the fixed effect model. More over, the hypothesis of equal preference functions for all groups cannot even be formulated. In the random effect model a preference function counting for each group has an interpretation as a conditional expected preference function and the hypothesis of equal preference functions counting for each group, can be tested by modifying the fixed effect test procedure.

Appendix

To construct a random effect model we follow Andersson (1990). For the present mean-value structure there is no difference between a random effect model and a canonical variance component model as defined by Andersson. All of the considered mean-values hypotheses are linear hypotheses and we consider the corresponding algebraic linear subspaces.

Only hypotheses/subspaces concerning preferences (θ_{ijp}) are of interest in the present appendix.

The Model (2) corresponds to $\theta_{ijp} = -\theta_{jip}$. The corresponding subspace is denoted L_0 . The hypothesis H_1 states that $\theta_{ijp} = \alpha_{jp} - \alpha_{ip}$ and the corresponding subspace is denoted L_1 , the hypothesis H_2 states that $\theta_{ijp} = \alpha_{j, \text{gender}(p)} - \alpha_{i, \text{gender}(p)}$ and the subspace is denoted L_2 .

The hypothesis H_3 states that $\theta_{ijp} = \alpha_j - \alpha_i$ and the subspace is denoted L_3 . The subspace corresponding to H_4 ($\alpha_1 = \dots = \alpha_6$) is the “zero-space”. Finally the sample space is denoted S , corresponding to the residual space.

Let L be the set of (geometrically orthogonal) subspaces $L = \{S, L_0, L_1, L_2, L_3\}$.

The random effect model can be represented by

$$Y = \sum_{B \in L} Y_B,$$

where Y_B , $B \in L$, is a family of independent stochastic variables, where Y_B is normally distributed on B with mean $\zeta \in M$ and variance $\sigma_B^2 P_B \cdot \delta$, where δ is the usual inner product restricted to B , P_B is the projection matrix onto B and $M = \Sigma(B' | B \supset B', B \neq B')$.

If Y_B is considered as a (degenerated) random variables on S (with support B) the covariance matrix becomes $\sigma_B^2 P_B$. The covariance matrix for Y becomes

$$\text{Cov}(Y) = \sum_{B \in L} \sigma_B^2 P_B.$$

The variance components, σ_B^2 , $B \in L$, can be estimated by the “usual” procedure by defining

$$\lambda_B = SS_B / df_B, \quad B \in L,$$

which are the unbiased estimates. The variance components is found by solving the equations

$$\lambda_B = \sum (\sigma_B^2 \cdot |B' \supseteq B).$$

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